ABSTRACT

Bring Your Own Vulnerable Driver (BYOVD) is a technique where an attacker loads a legitimate but exploitable kernel driver to gain elevated privileges on Windows systems. To counter this, modern Endpoint Detection and Response (EDR) agents deploy telemetry gathering in kernel space. However, attackers increasingly use BYOVD primitives to terminate or blind these user-mode analysis services. This paper presents a comprehensive, lightweight detection and self-protection framework. We implement the core logic in kernel mode as a hybrid Windows Driver Model (WDM) and Minifilter driver that intercepts driver loads in real-time while actively stripping dangerous access rights (such as termination and virtual memory writing) from any handles opened against the associated user-mode service. The user-mode engine, developed in C# on the .NET framework, receives telemetry via a Filter Manager communication port, performs offline hash attestation against a curated blocklist of known vulnerable drivers, and exposes a real-time web dashboard for analyst decision-making. By delegating disk I/O and cryptographic hashing to user-mode while maintaining strict self-defense in Ring 0, the architecture ensures kernel stability, prevents EDR blinding, and imposes minimal execution overhead on the critical image-load path.